SOLVED PROBLEMS

Quantum Mechanics & Applications – CSE Stream [BPHYS102/202]

Module-5 : Quantum Computation

Syllabus: Numerical Problems on - Prediction of Circuits with single/two qubit gates, Unitary matrices & Matrix operations.

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1. Find the norm of $|U\rangle = \begin{bmatrix} 1-i \\ 2 \end{bmatrix}$

Given: $|U\rangle = \begin{bmatrix} 1-i \\ 2 \end{bmatrix}$

$\therefore \langle U| = |U\rangle^{*T} = [1+i \quad 2]$.

Norm of $|U\rangle = \langle U|U\rangle = [1+i \quad 2] \begin{bmatrix} 1-i \\ 2 \end{bmatrix} = 1^2 - i^2 + 4 = 1 + 1 + 4$

$\langle U|U\rangle = 6$

2. Find the inner product of $|A\rangle = \begin{bmatrix} a \\ ib \end{bmatrix}$ with itself.

Given: $|A\rangle = \begin{bmatrix} a \\ ib \end{bmatrix}$

$\therefore \langle A| = |A\rangle^{*T} = [a \quad -ib]$

Inner product $\langle A|A\rangle = [a \quad -ib] \begin{bmatrix} a \\ ib \end{bmatrix} = a^2 - i^2 b^2 = a^2 + b^2$

$\langle A|A\rangle = a^2 + b^2$

3. If $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, prove that $A^{\dagger} = A$. (JUN / JUL - 2023)

If $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, then $A^{*} = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$

Then $A^{*T} = A^{\dagger} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

$\therefore \quad A^{\dagger} = A$

4. Find the inner product of the states $|0\rangle$ & $|1\rangle$ and draw conclusions on the result. (JUN / JUL - 2023)

We have $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\langle 0| = [1\ 0]$ and $\langle 1| = [0\ \ 1]$.

$\langle 0|0\rangle = [1\ \ 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 + 0 = 1$ So, $|0\rangle$ is normalized

$\langle 1|1\rangle = [0\ \ 1] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 + 1 = 1$ So, $|1\rangle$ is also normalized

$\langle 0|1\rangle = [1\ \ 0] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 + 0 = 0$ and $\langle 1|0\rangle = [0\ \ 1] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 + 0 = 0$

So, $|0\rangle$ and $|1\rangle$ are orthogonal to each other.

Hence the set of vectors $|0\rangle$ and $|1\rangle$ is orthonormal.

The physical significance of the orthonormal condition is that if $|0\rangle$ and $|1\rangle$ represent the two spin states of a particle, then the spins are mutually exclusive / independent.

5. Explain the matrix representation of $|0\rangle$ & $|1\rangle$ states. Apply identity operator on them.

(DEC-2023 / JAN-2024 & Model QP-2)

The matrix representations are $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

We have $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

There will be no change when it operates on $|0\rangle$ or $|1\rangle$

$$I|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{That is,} \quad I|0\rangle = |0\rangle$$

$$I|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{That is,} \quad I|1\rangle = |1\rangle$$

6. Given $|\psi\rangle = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$ & $|\phi\rangle = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$. Prove that $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$.

(Model QP-2, DEC-2024 / JAN-2025, JUN/JULY - 2025 & DEC-2025 / JAN-2026)

We can write $\langle\psi| = [\alpha_1^* \alpha_2^*]$ & $\langle\psi| = [\beta_1^* \beta_2^*]$

$$\therefore \langle\psi|\phi\rangle = \begin{bmatrix} \alpha_1^* \alpha_2^* \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \alpha_1^* \beta_1 + \alpha_2^* \beta_2$$

$$\& \langle\phi|\psi\rangle = \begin{bmatrix} \beta_1^* \beta_2^* \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \alpha_1 \beta_1^* + \alpha_2 \beta_2^*$$ Or

$$\langle\phi|\psi\rangle^* = \alpha_1^* \beta_1 + \alpha_2^* \beta_2$$

$$\therefore \langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$$

7. A linear operator X operates such that $X|0\rangle = |1\rangle$ & $X|1\rangle = |0\rangle$. Find the matrix representation of X.
(DEC-2024 / JAN-2025, DEC-2023&JAN-2024, JAN/ FEB -2023 &Model QP-1)

We have $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Let $X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$

Given $X|0\rangle = |1\rangle$

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad => \quad x_{11} = 0 \quad \& \quad x_{21} = 1$$

Also, $X|1\rangle = |0\rangle$

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad => \quad x_{12} = 1 \quad \& \quad x_{22} = 0$$

$$\therefore X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

8. A linear operator Y operates such that $Y|0\rangle = i|1\rangle$ & $Y|1\rangle = -i|0\rangle$. Find the matrix representation of Y.

We have $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Let $Y = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$

Given $Y|0\rangle = i|1\rangle$

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} y_{11} \\ y_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} \quad => \quad y_{11} = 0 \quad \& \quad y_{21} = i$$

Also, $Y|1\rangle = -i|0\rangle$

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} y_{12} \\ y_{22} \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} \quad => \quad y_{12} = -i \quad \& \quad y_{22} = 0$$

$$\therefore Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

9. A linear operator Z operates such that $Z|0\rangle = |0\rangle$ & $Z|1\rangle = -|1\rangle$. Find the matrix representation of *Z*.

We have $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Let $Z = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$

Given $Z|0\rangle = |0\rangle$

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} z_{11} \\ z_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad => \quad z_{11} = 1 \quad \& \quad z_{21} = 0$$

Also, $Z|1\rangle = -|1\rangle$

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} z_{12} \\ z_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad => \quad z_{12} = 0 \quad \& \quad z_{22} = -1$$

$$\therefore Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

10. Using matrix multiplication, show that on applying Hadamard gate twice to a $|0\rangle$ vector results in its original state.

We have $H = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$

$$\therefore HH = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1+1 & 1-1\\ 1-1 & 1+1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

Now, consider $HH|0\rangle = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix}$

$$\therefore HH|0\rangle = |0\rangle$$

11. Show that Hadamard gate is unitary.

We have $H = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$

$$\therefore \quad H^{*} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$$

Then $H^{\dagger} = H^{*T} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$

$$\therefore HH^{\dagger} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1+1 & 1-1\\ 1-1 & 1+1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

Or $HH^{\dagger} = I$

Hence Hadamard gate is unitary.

12. Using two X gates in series form a quantum wire.

The required condition here is XX = I

We have $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Then $XX = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\therefore XX = I$

13. Show that *S* gate can be formed by connecting two T gates in series. (DEC-2025/JAN-2026)

We have $T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$ and $S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

Consider $TT = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

$\therefore T^2 = S$

[Note: $e^{i\pi/2} = \cos \pi/2 + i \sin \pi/2 = 0 + i \times 1 = i$]

14. Show that the matrix $A = \begin{bmatrix} i & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & i \end{bmatrix}$ is unitary.

Given matrix $A = \begin{bmatrix} i & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & i \end{bmatrix}$

Then, its conjugate transpose = $A^{*T} = A^{\dagger} = \begin{bmatrix} -i & 0 & 0 \\ 0 & -i & 0 \\ 0 & 0 & -i \end{bmatrix}$

$$AA^{\dagger} = \begin{bmatrix} i & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & i \end{bmatrix} \times \begin{bmatrix} -i & 0 & 0 \\ 0 & -i & 0 \\ 0 & 0 & -i \end{bmatrix}$$

$$AA^{\dagger} = \begin{bmatrix} -i^2 & 0 & 0 \\ 0 & -i^2 & 0 \\ 0 & 0 & -i^2 \end{bmatrix}$$

We have $i^{-2} = -1$ or $-i^{-2} = 1$

$$\therefore AA^{\dagger} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \text{Identity matrix} = I$$

Since $AA^{\dagger} = I$, matrix A is unitary.

15. Show the action of Hadamard gate on of $|0\rangle$ & $|1\rangle$ states.

H- gate is given by the matrix $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|1\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

16. A quantum state is given by $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \frac{\sqrt{2}}{\sqrt{3}}|1\rangle$ Calculate the probability of obtaining $|1\rangle$.

Probability of obtaining $|1\rangle$ is $P^{|1\rangle} = |\langle 1\|\psi\rangle|^2$

$$P^{|1\rangle} = \left|\langle 1\left\| \left(\frac{1}{\sqrt{3}}|0\rangle + \frac{\sqrt{2}}{\sqrt{3}}|1\rangle\right)\right|\right|^2$$

$$P^{|1\rangle} = \left|\frac{1}{\sqrt{3}}\langle 1\|0\rangle + \frac{\sqrt{2}}{\sqrt{3}}\langle 1\|1\rangle\right|^2$$

$$P^{|1\rangle} = \left|\frac{1}{\sqrt{3}} \times 0 + \frac{\sqrt{2}}{\sqrt{3}} \times 1\right|^2$$

$$P^{|1\rangle} = \frac{2}{3} = 0.67$$